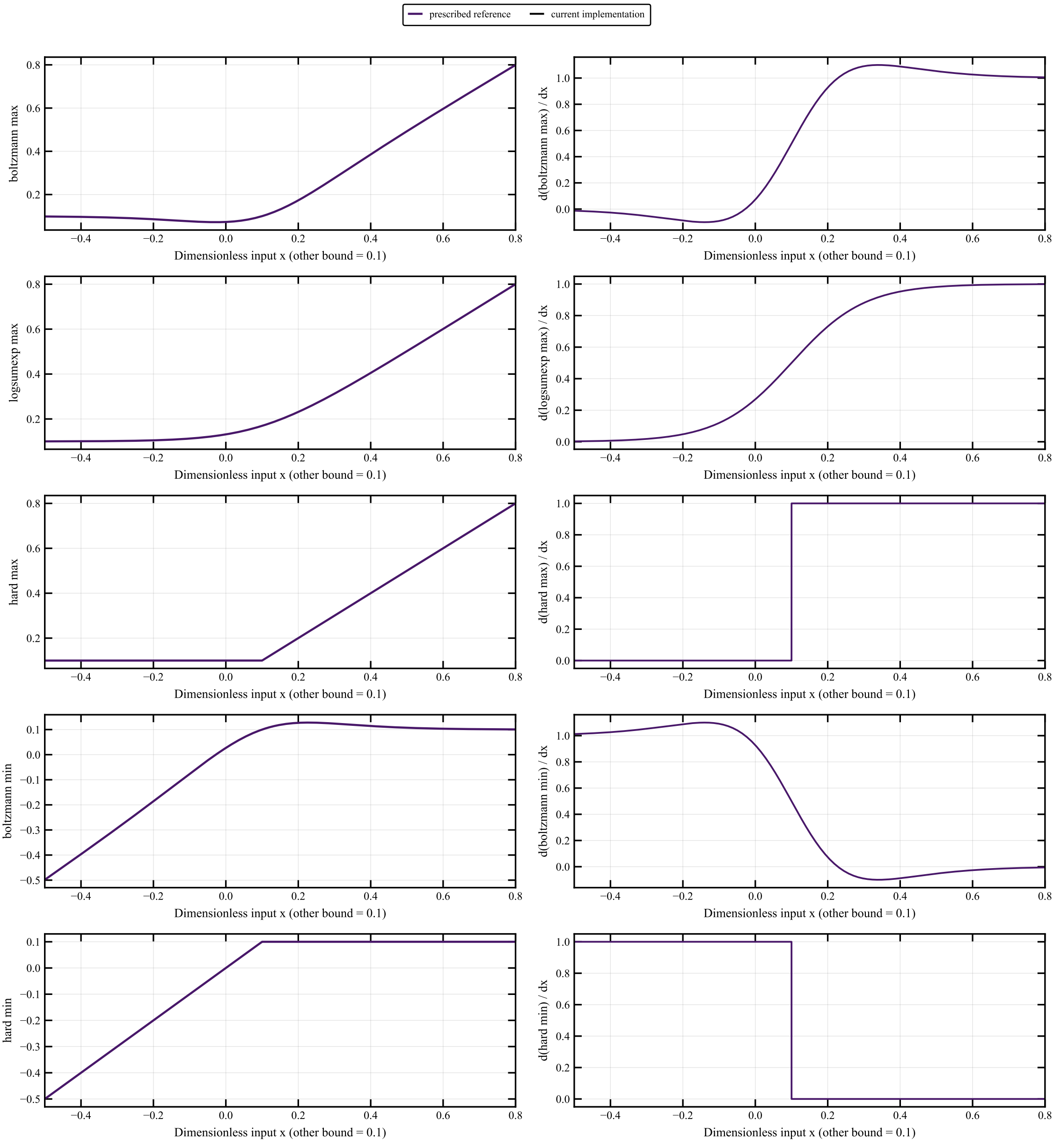smooth bounds — values and first derivatives